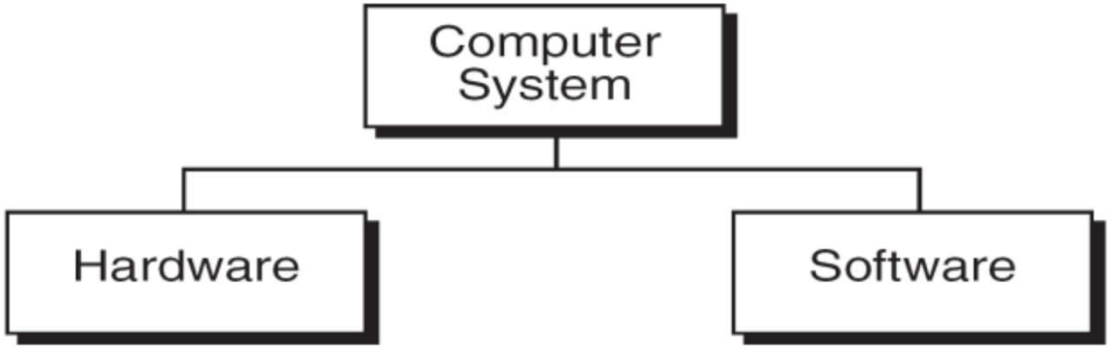
BAEKSEOK UNIVERSITY

C Intro.

01 Computer System

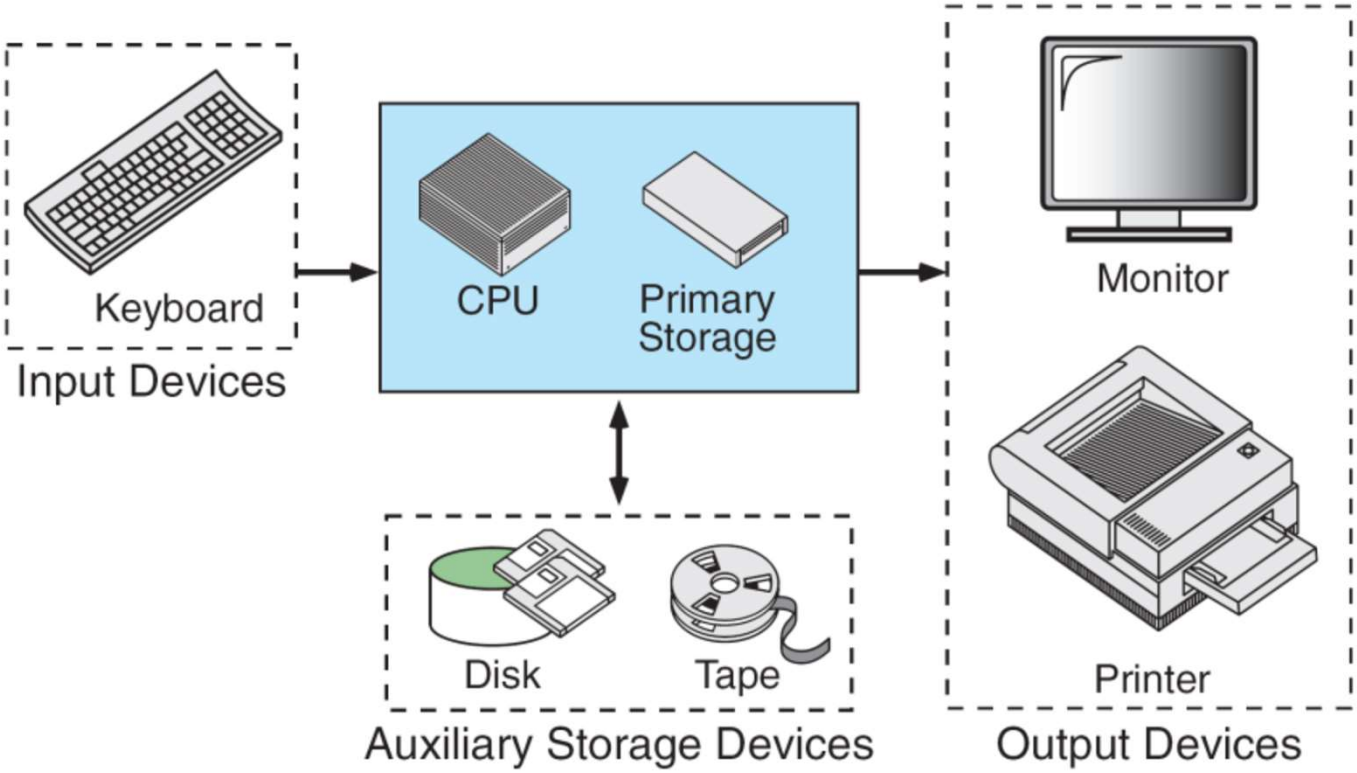
Computer System

- A Computer is an electronic device which performs operations such as accepts data
- As an input, store the data, manipulate or process the data and produce the results an output.
- Main task performed by a computer
  - Accept the data
  - Process or manipulate the data
  - Display or store the result in the form of human understanding
  - Store the data, instructions and results.



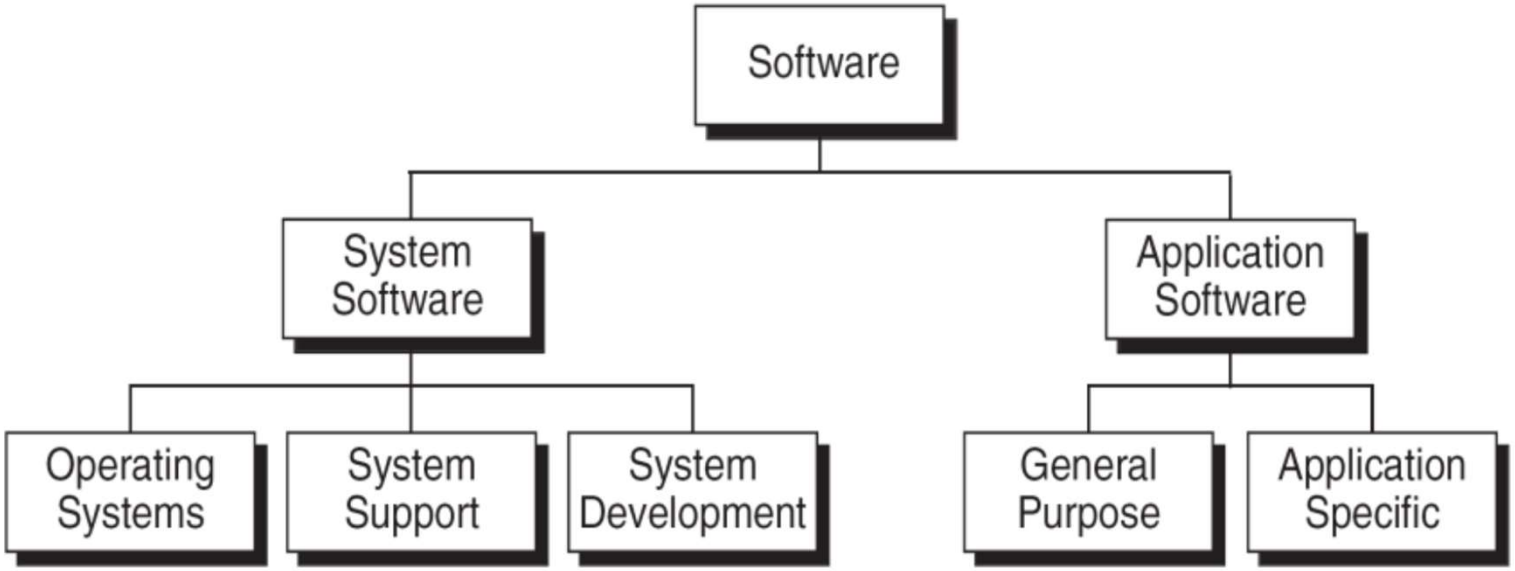
Computer System

- Computer hardware is the collection of physical elements that comprise a computer system



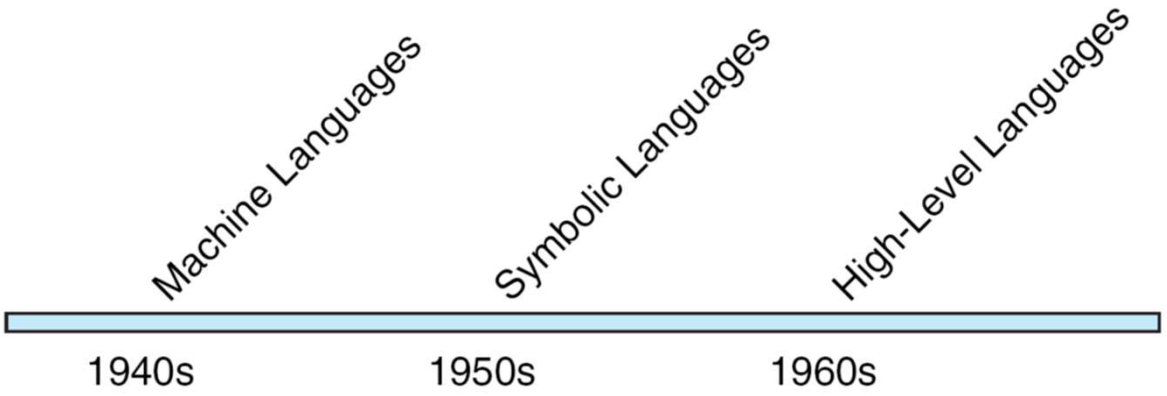
Computer System

- Computer software is a collection of computer programs and related data that provides the instructions for a computer what to do and how to do it. Software refers to one or more computer programs and data held in the storage of the computer for some purpose



COMPUTER LANGUAGES

- In order to communicate with the computer user also needs to have a language that should be understood by the computer.
- For this purpose, different languages are developed for performing different types of work on the computer.
- Basically, languages are divided into two categories according to their interpretation.



High Level Languages

a) Algebraic Formula-Type Processing.

- BASIC (Beginners All Purpose Symbolic Instruction Code).
- FORTRAN (Formula Translation).
- PL/I (Programming Language, Version 1).
- ALGOL (Algorithmic Language).
- APL (A Programming Language).

b) Business Data Processing:

- COBOL (Common Business Oriented Language).
- RPG (Report Program Generator)

c) **String and List Processing**

- LISP (List Processing).
- Prolog (Program in Logic).

d) Object Oriented Programming

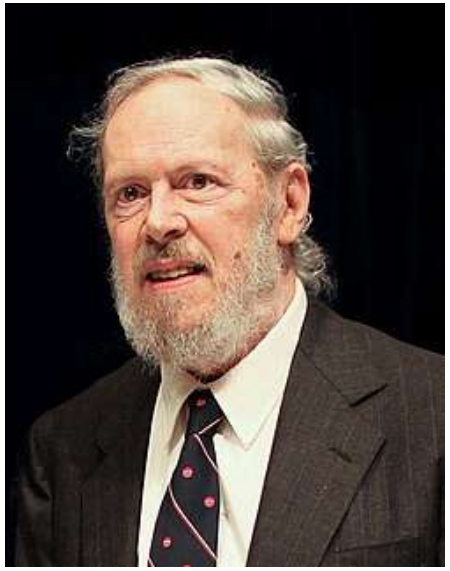
- C++
- Java

e) Visual programming language:

- Visual Basic
- Visual Java
- Visual C

What is C?

- C is a general-purpose programming language created by Dennis Ritchie at Bell Labs in 1972.
- It is popular because it is fast, widely supported, and helps you understand how programs work "under the hood".
- The main reason for its popularity is because it is a fundamental language in the field of computer science.
- C is closely connected to UNIX, because much of UNIX was written in C.



What is C used for?

- C is often used to create programs that need to run fast and work closely with the computer.
  - Operating systems, such as parts of Windows, Linux, and macOS
  - Programs inside devices like cars, TVs, and home electronics
  - Software that needs to be fast, including databases and system tools
  - Game engines and programs that handle graphics
  - Core libraries that other programming languages rely on

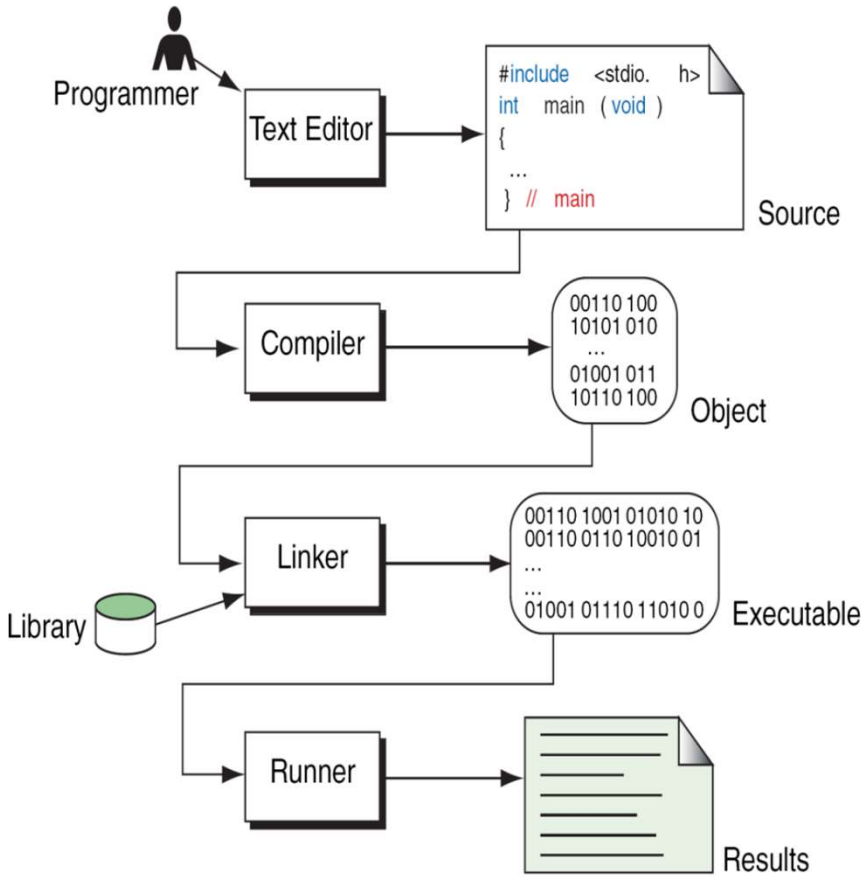
Why Learn C?

- It is one of the most widely used programming languages
- If you know C, you will have no problem learning other popular programming languages such as Java, Python, C++, C#, etc, as the syntax is similar
- It helps you understand memory, performance, and how computers handle data
- C is very versatile; it can be used in both applications and technologies

Programming

- Createing and Running Programs:

1. Writing and editing the program using Text editor (source code).
2. Compile the program using any C compiler.(.bak file)
3. Linking the program with the required library modules(object file)
4. Executing the program. (.Exe file)



Algorithm

- An algorithm is a description of a procedure which terminates with a result. Algorithm is a step-by-step method of solving a problem.
- Properties of an Algorithm:

1) *Finiteness*: - An algorithm terminates after a finite numbers of steps.
2) *Definiteness*: - Each step in algorithm is unambiguous. This means that the action specified by the step cannot be interpreted (explain the meaning of) in multiple ways & can be performed without any confusion.
3) *Input*: - An algorithm accepts zero or more inputs
4) *Output*:- An algorithm should produce at least one output.
5) *Effectiveness*: - It consists of basic instructions that are realizable. This means that the instructions can be performed by using the given inputs in a finite amount of time.

Algorithm

- Writing an algorithm
  - An algorithm can be written in English, like sentences and using mathematical formulas.
  - Sometimes algorithm written in English like language is Pseudo code.
  - Examples
    - Finding the average of three numbers

1) Let a,b,c are three integers
2) Let d is float
3) Display the message "Enter any three integers:"
4) Read three integers and stores in a,b,c
5) Compute the d = (a+b+c)/3.0
6) Display "The avg is:"
7) End.

Pseudocode::

- **Input a set of 4 marks**
- **Calculate their average by summing and dividing by 4**
- **if average is below 50**
  - **Print "FAIL"**
- **else**
  - **Print "PASS"**

- **Detailed Algorithm :**

```
Step 1: Input M1,M2,M3,M4
Step 2: GRADE <- (M1+M2+M3+M4)/4
Step 3: if (GRADE < 50) then
            Print "FAIL"
        else
            Print "PASS"
        endif
```

Requirement

- Build tool
  - Compiler
  - Linker
  - Editor

- IDE(Integrated Develop Environment)
  - Visual studio
  - Visual Studio code + others
  - Eclipse
  - Etc…

- hello.c

```c
#include <stdio.h>

int main() {
  printf("Hello World!");
  return 0;
}
```

BAEKSEOK UNIVERSITY

Thank You !

Copyright 2026
All rights reserved.